

Fundamental Frequency Extraction Using DCT Based Cumulative Power Spectrum in Noisy Speech

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Outlines

- ☐ Introduction
- ☐ Pitch Detection Methods
- ☐ Research Objective & Motivation
- ☐ Proposed Method
- ☐ Experimental Results
- ☐ Conclusion

Introduction

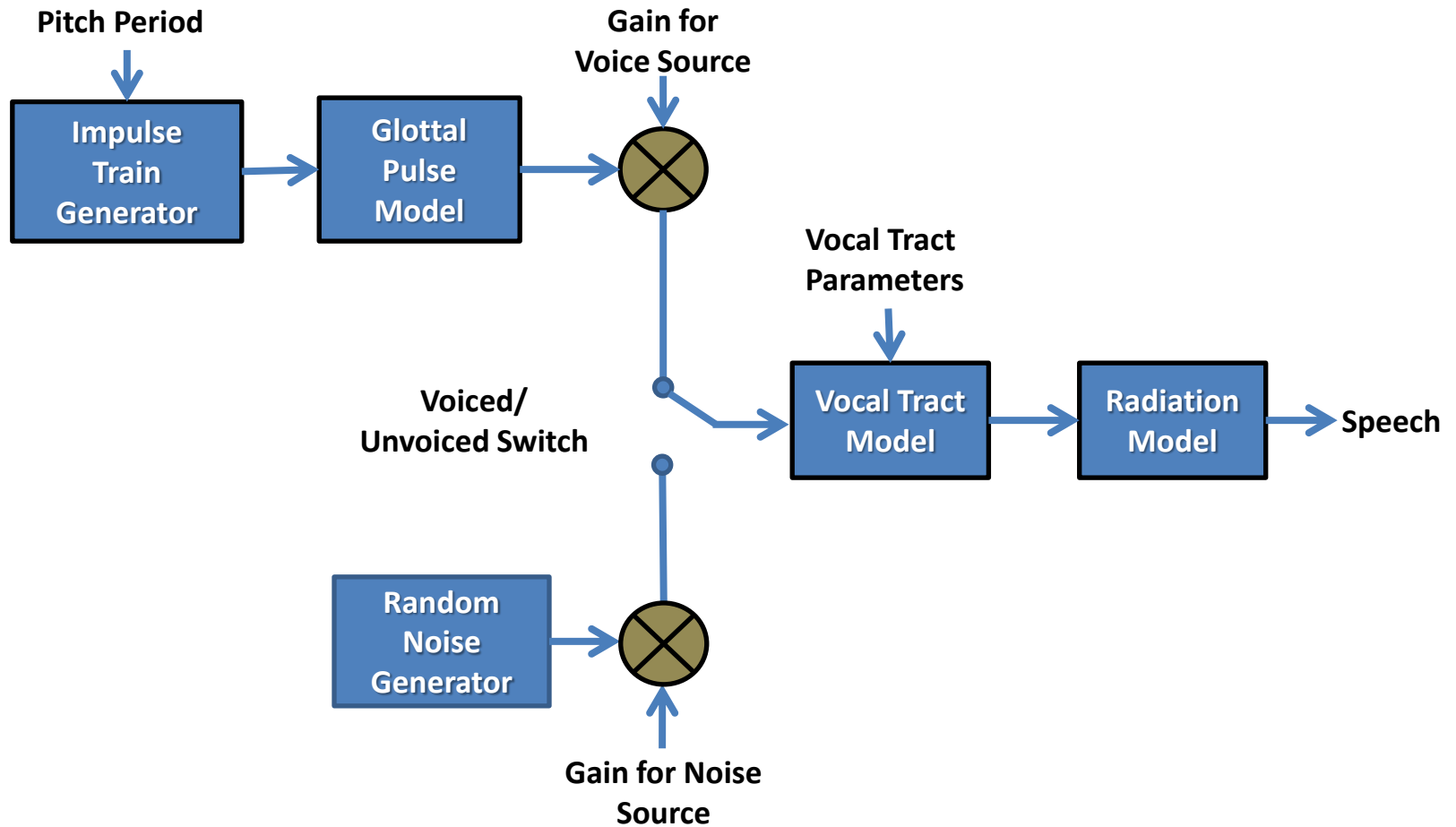


Figure 1: Speech Production Model

Speech Signal

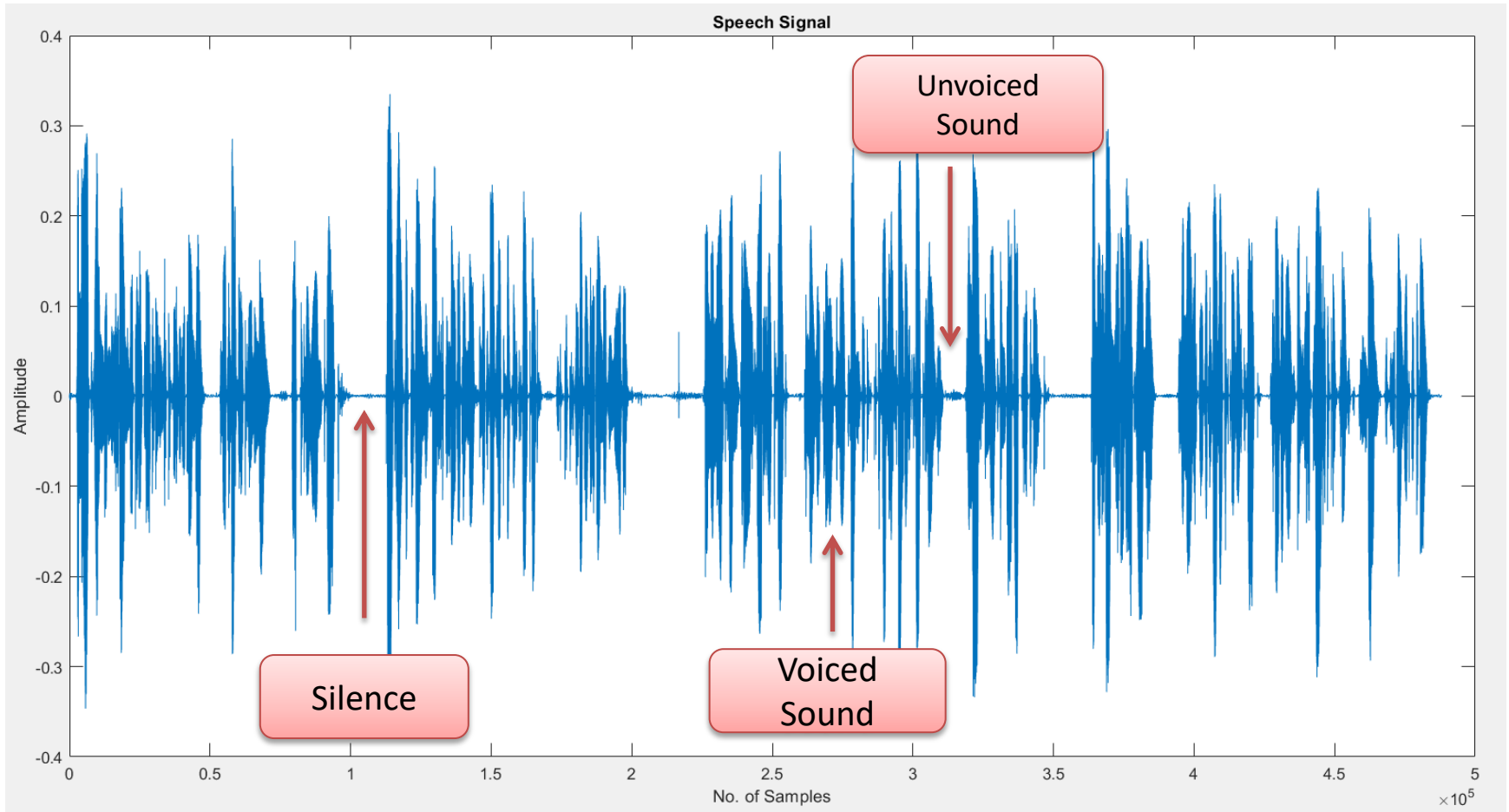
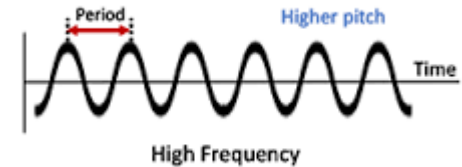
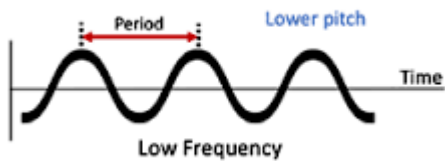
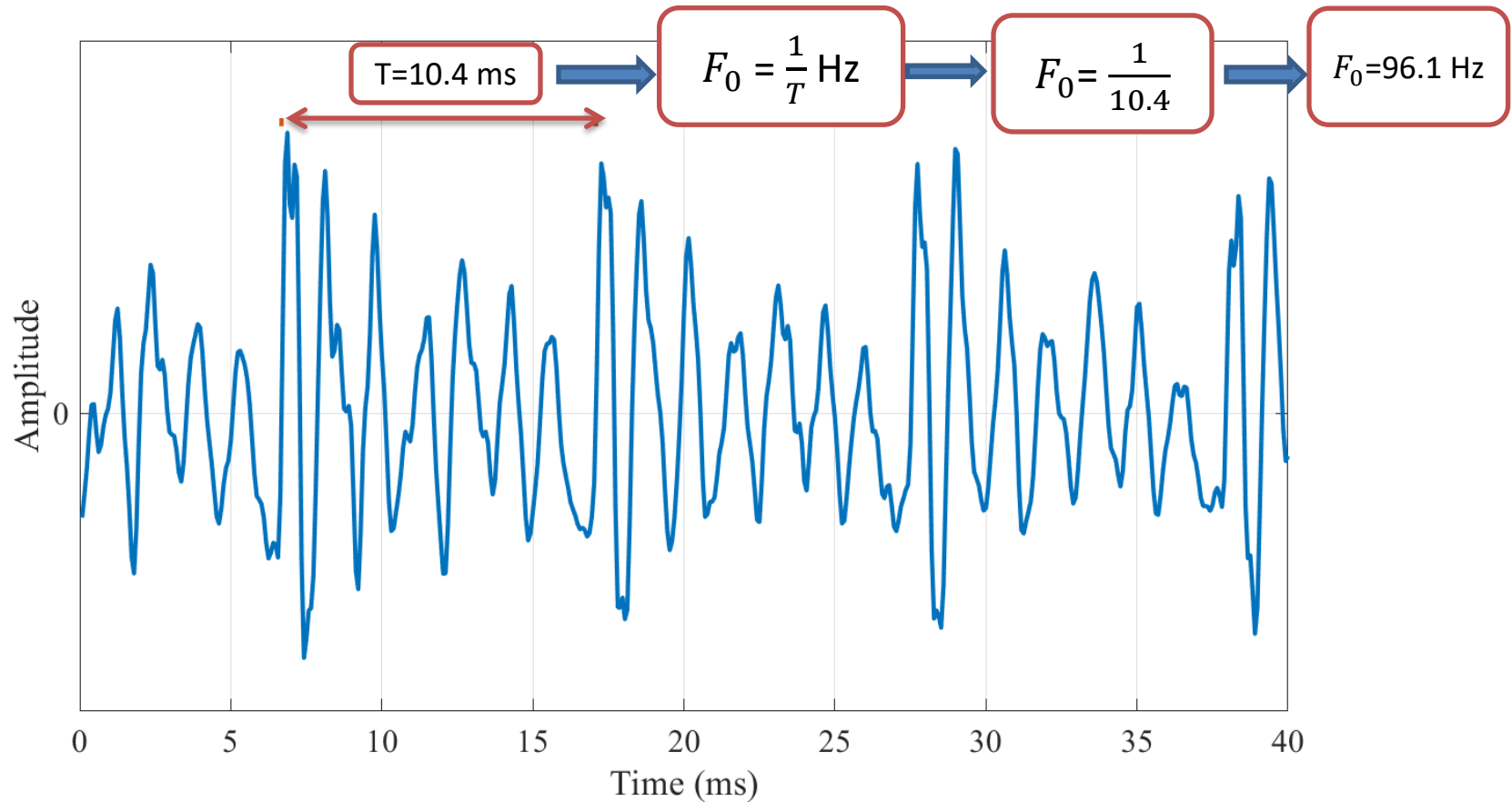


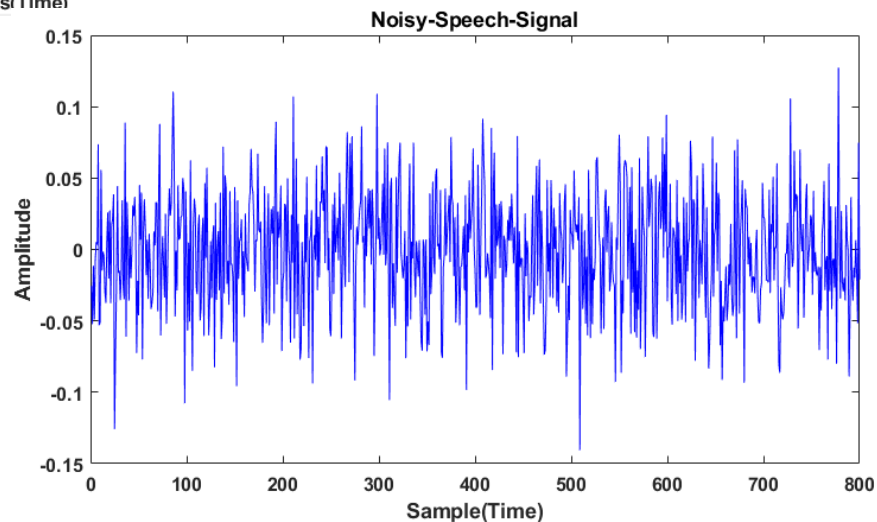
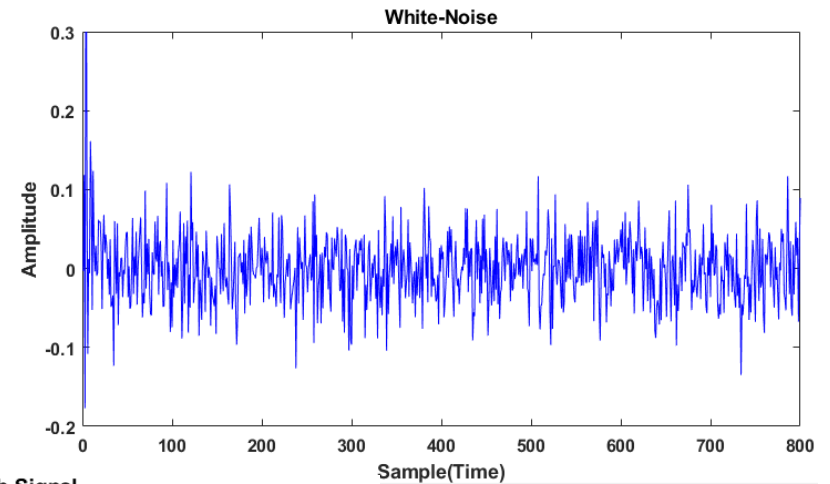
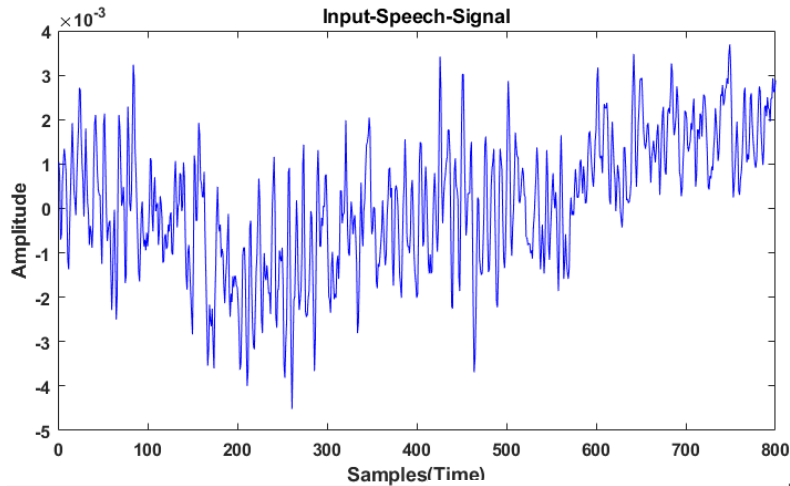
Figure 2: States of Speech Signal

Fundamental Frequency



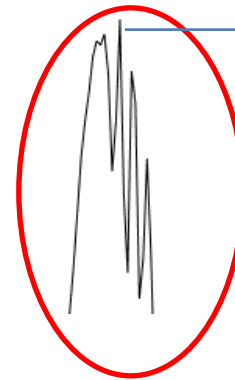
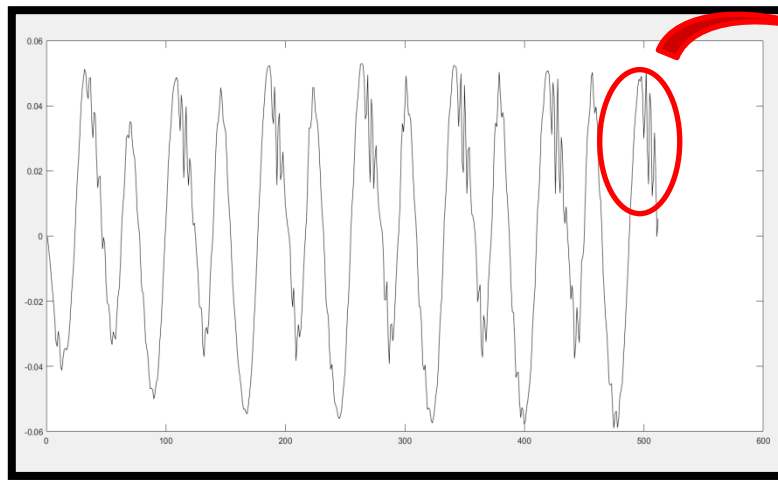
Difficulties in Extracting Pitch

Background noise



Difficulties in Extracting Pitch(Contd.)

Vocal Tract Effect

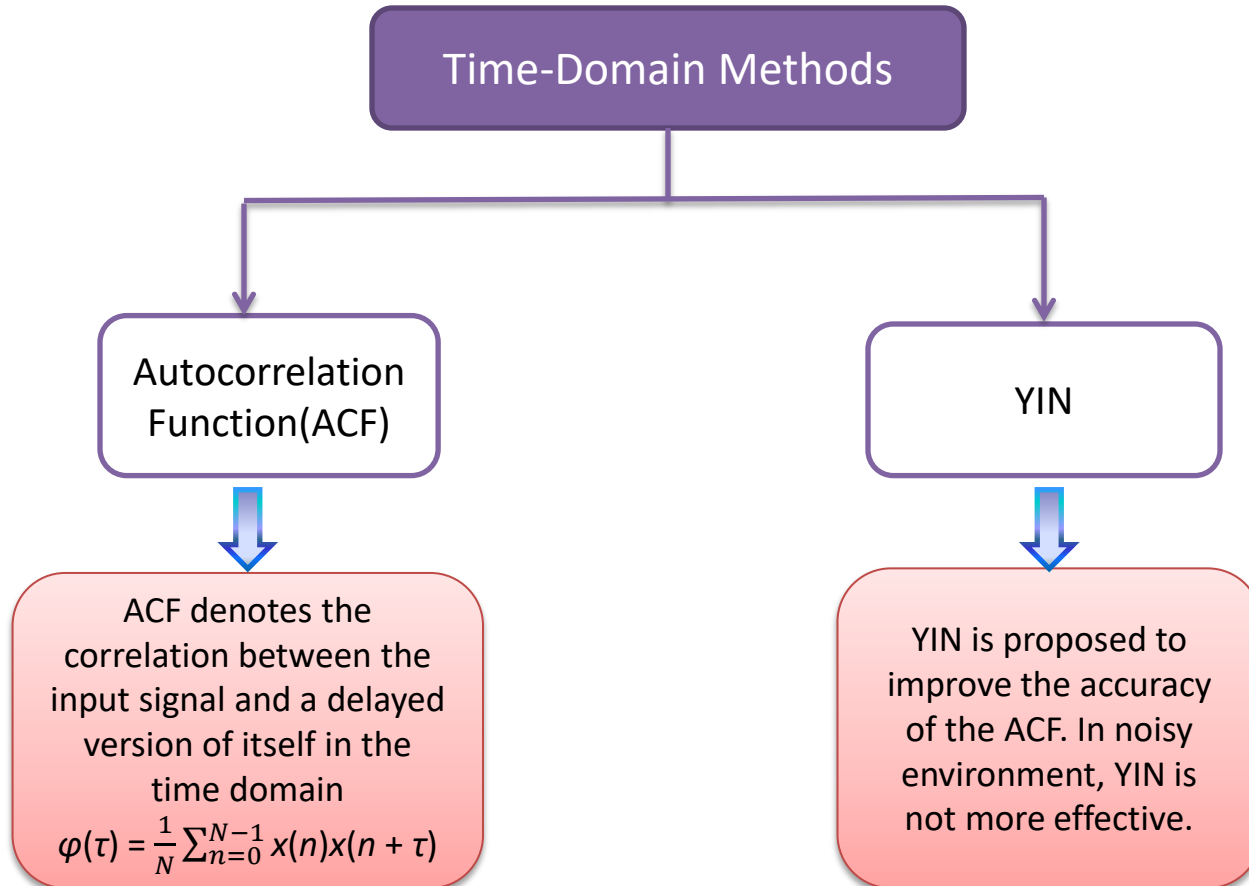


The fluctuation in amplitude or frequency.

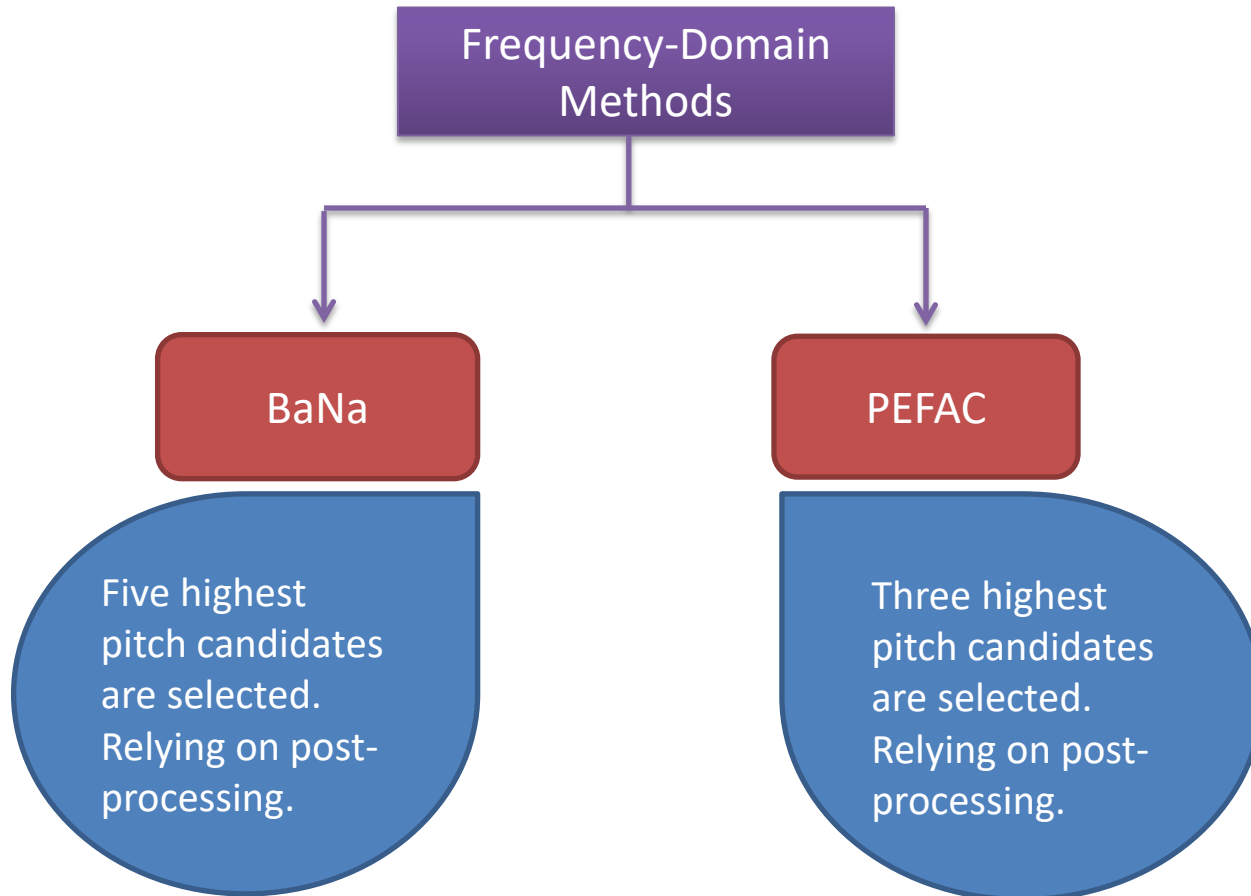
Ripple

We need to remove the vocal tract filter for better pitch extraction.

Related Works



Related Works (Contd.)

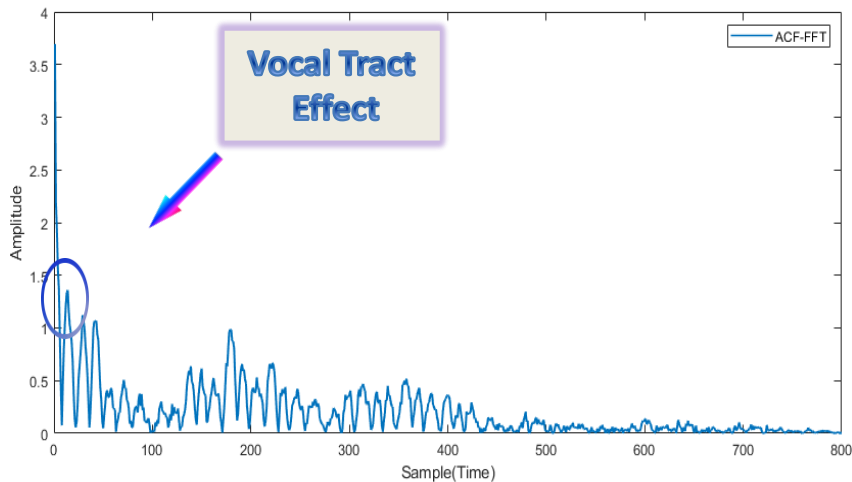


Objectives

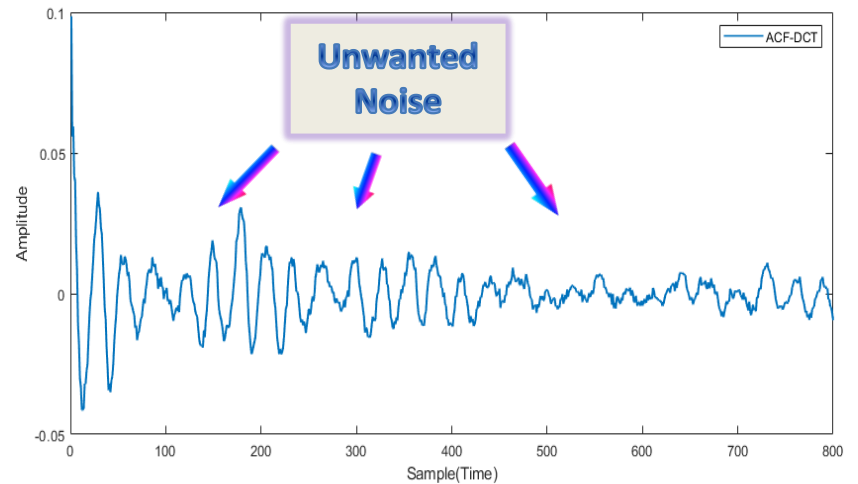
Design a pitch extraction algorithm which is:

- Simple and efficient.
- Time saving.
- Robust against both noise and vocal tract effect.
- Utilized without relying on post-processing.
- Achieved more smooth signal and better accuracy.
- Overcomes the struggling of other methods within low SNR.

Motivation



In a search to reduce vocal tract effect that the conventional ACF can't overcome.



In an aim to achieving more smooth signal.

Proposed Method

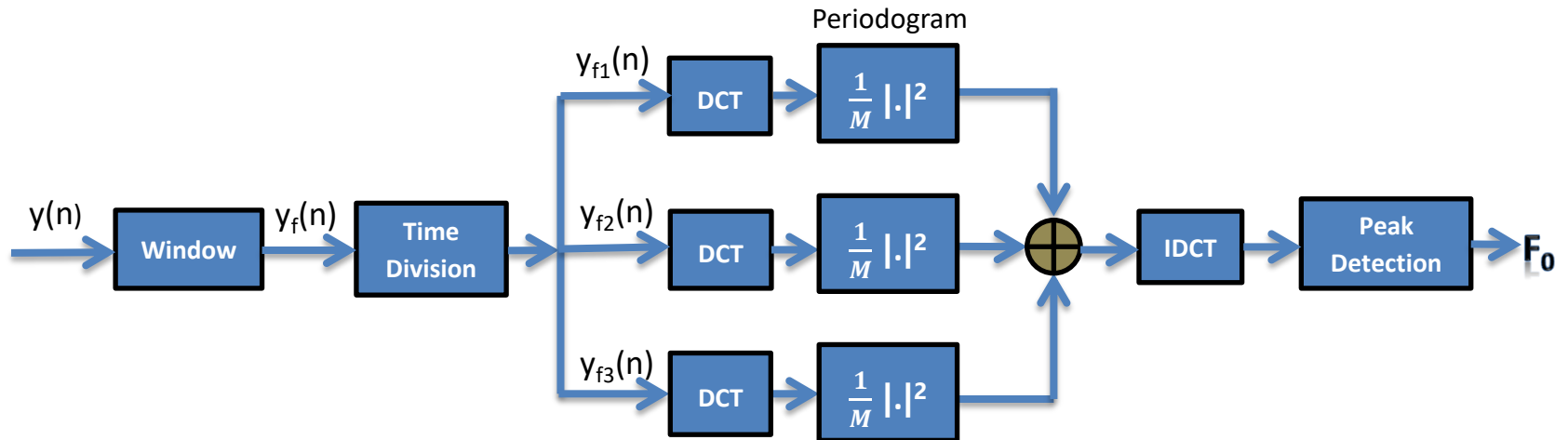


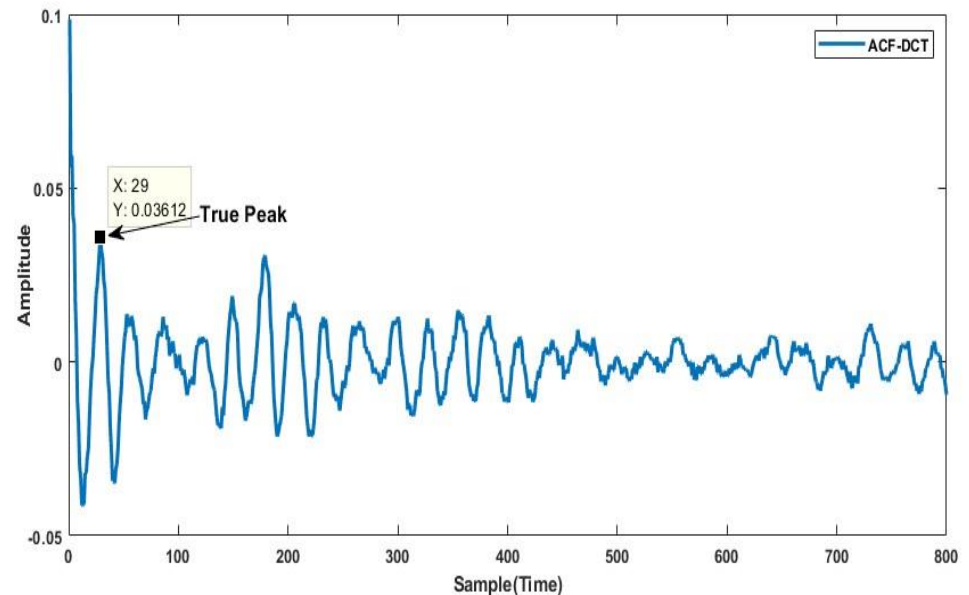
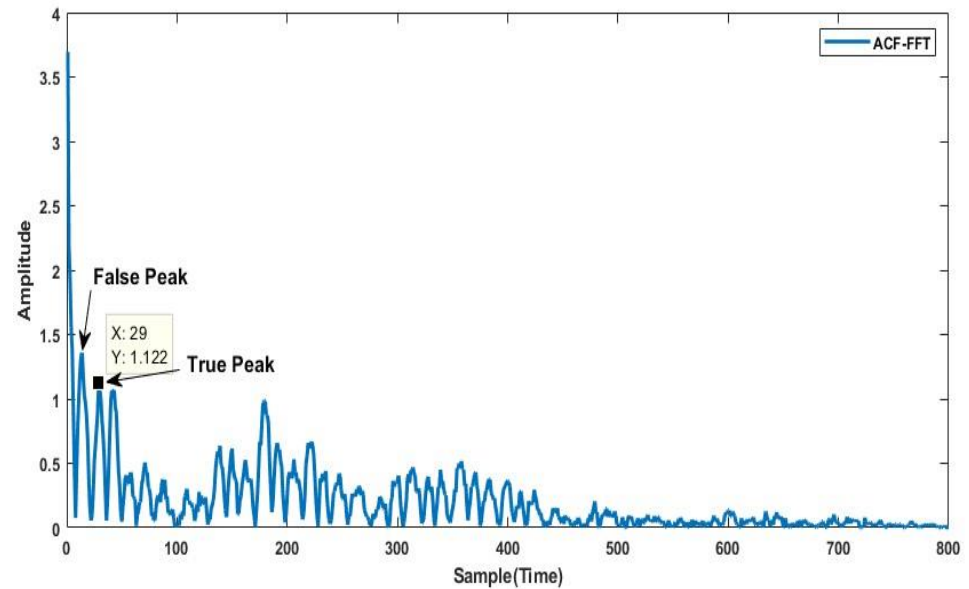
Figure 3: Block Diagram of DCT Based Cumulative Power Spectrum

Proposed Method(Contd.)

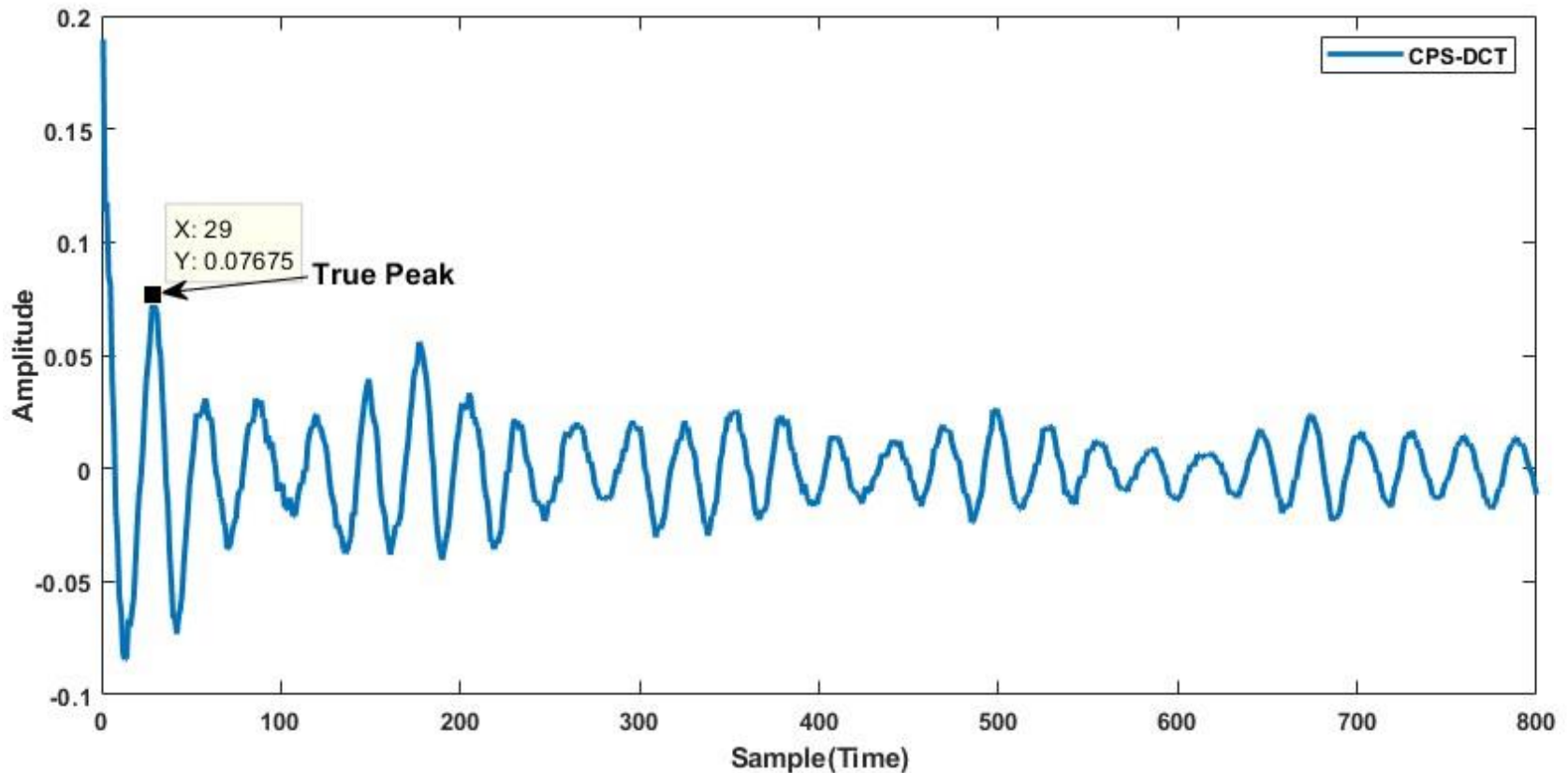
Frame Length : 50 ms
Sampling Frequency : 16000 Hz

Number of Samples Per Frame Length

$$= \frac{50 \text{ ms}}{1000} * 16000 \text{ Hz}$$
$$= 800 \text{ Samples}$$



Proposed Method (Contd.)



- + Reduce the unsmooth occurrences due to Vocal Tract Effect .
- + More Smooth Signal is achieved by reducing noise characteristics.

Experimental Conditions

Table 1: Experimental Conditions

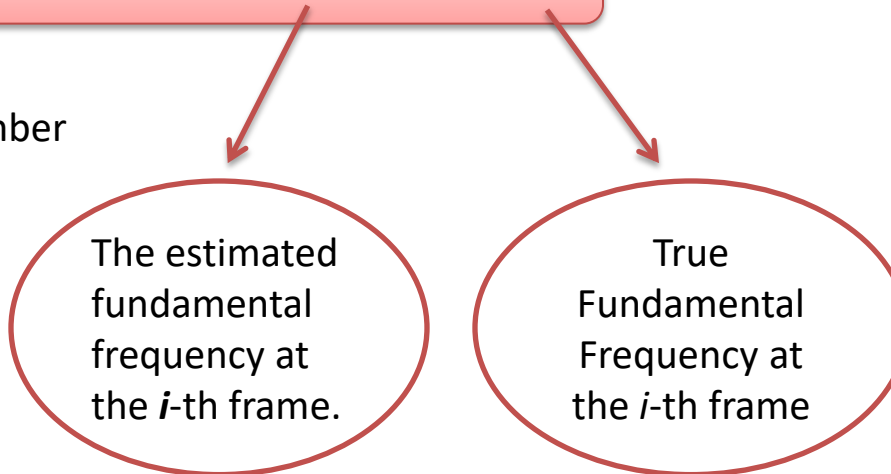
Experimental Data	KEELE Database (Five Male Speech and Five Female Speech).
Sampling frequency	16 [kHz]
Frame length	50 [ms] except for PEFAC (90ms) and BaNa (60 ms).
Frame shift	10 [ms].
DCT (IDCT) Points	2048 points except for PEFAC (2^{13} points) and BaNa (2^{16} points)
Window function	Rectangular except for PEFAC (Hamming) and BaNa (Hanning).
Noise type	White Noise, Babble Noise, Train Noise, HF-Channel Noise.
SNR level	-5, 0, 5, 10, 15, 20 [dB]

Evaluation Criteria

Pitch estimation error is determined by measuring the difference between the reference and estimated fundamental frequencies .

Based on Rabinar's rule: $e(i) = F_{est}(i) - F_{true}(i)$

Here,
i = Frame Number



If $|e(i)| > 10\%$ of the $F_{true}(i)$ then, it is called **Gross Pitch Error (GPE)**.
Otherwise, it is called **Fine Pitch Error**.

Experimental Results

Table 2: Average GPE Error Rate for White Noise

SNR	CPS-DCT(Proposed)	PEFAC	BaNa	YIN
-5	29.55031	41.04161	27.44827	48.99407
0	20.58926	37.15398	22.61711	31.3701
5	15.9896	34.38029	19.58053	21.59396
10	13.86573	33.01514	17.80155	16.57429
15	13.12183	32.50849	16.97529	14.29189
20	12.90138	31.98937	16.59966	12.8749

Table 3: Average GPE Error for Babble Noise

SNR	CPS-DCT(Proposed)	PEFAC	BaNa	YIN
-5	52.91545	58.59012	55.15213	52.5633
0	35.182864	49.01548	40.54781	36.8909
5	22.888482	41.86631	29.48598	23.68052
10	16.577184	37.4189	22.84607	16.64691
15	13.09545	34.98688	19.69209	13.1676
20	11.875954	33.395	17.70592	12.14275

Experimental Results (Contd.)

Table 4: Average GPE Error Rate for Train Noise

SNR	CPS-DCT(Proposed)	PEFAC	BaNa	YIN
-5	49.12419	50.73216	39.07639	50.23369
0	33.44969	43.17706	29.08431	34.38243
5	22.81971	38.9921	23.11176	22.76062
10	16.98957	35.59503	20.04322	16.36671
15	14.50046	33.40132	18.31063	13.42459
20	13.49092	32.25406	17.36339	12.16158

Table 5: Average GPE Error for HF-Channel Noise

SNR	CPS-DCT(Proposed)	PEFAC	BaNa	YIN
-5	38.53928	45.69573	27.1985	49.37144
0	24.70326	40.13406	22.64961	31.55247
5	17.64917	36.86074	19.82379	21.01063
10	14.79154	34.37672	17.90747	16.06053
15	13.45732	32.98622	17.31619	13.76763
20	13.04867	32.11695	16.57342	12.79615

Experimental Results (Contd.)

Observations

- Our suggested approach consistently demonstrates the lowest average GPE rate among all compared techniques across especially low SNRs in presence of Babble noise .
- No time consuming and complicated post-processing of the signal waveform is needed.
- Achieve more smooth signal and overcome the vocal tract effect efficiently by equalizing unnecessary ripples in the waveform .

Conclusion

The proposed method can be used for estimating accurate pitch from noisy signals with a higher accuracy by reducing the effects of noise characteristics and vocal tract.



THANK
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